

Engineering the drivetrain of a F24 car

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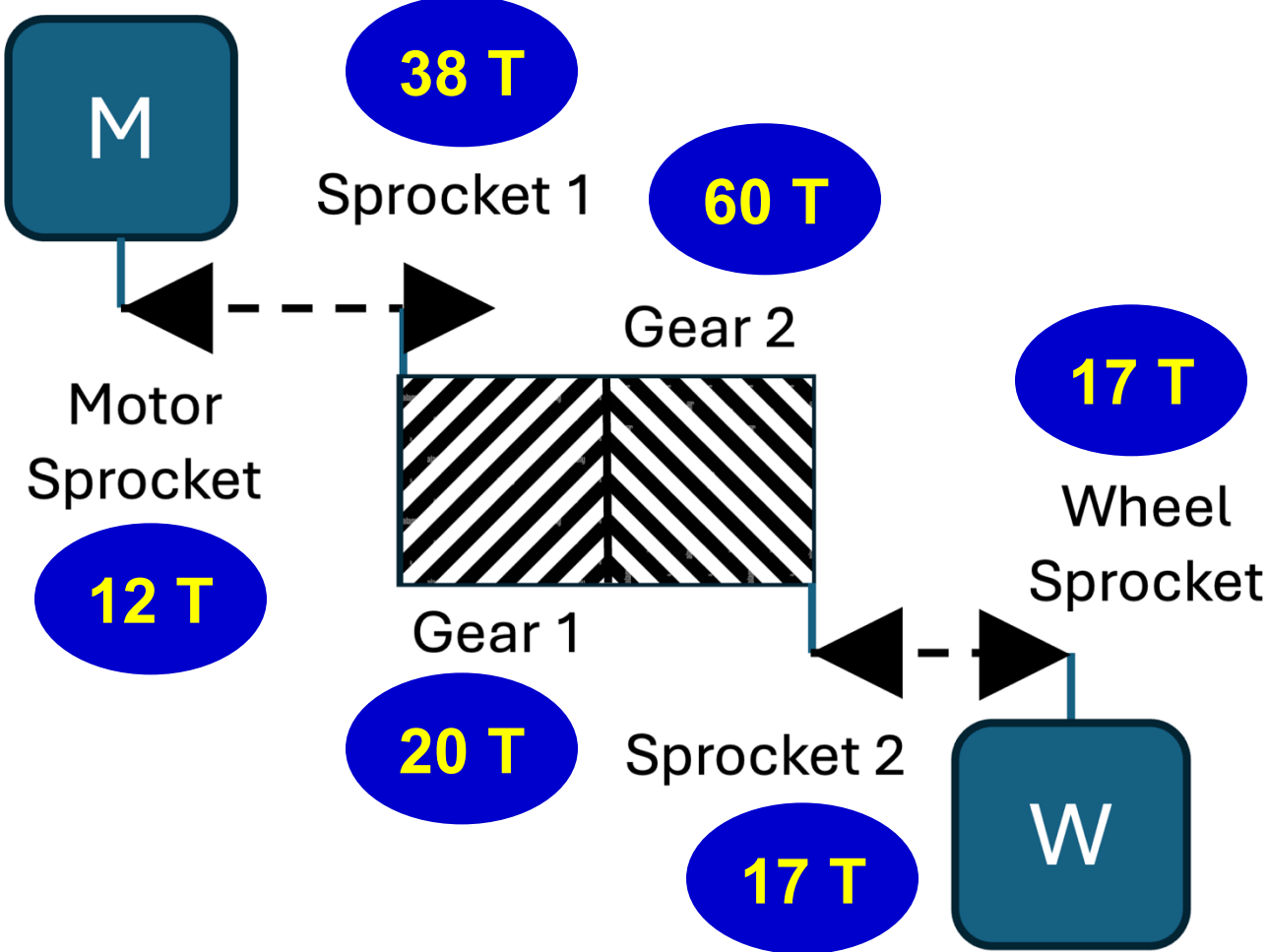
Group
P30

Background:

Project consists of designing and manufacturing a **drivetrain** connecting the **electric motor** and **the rear axle** of a F24 kit car.

- Components made or purchased on **RS Orders**
- Drivetrain fully enclosed within vehicle body
- Maximum top speed of **25 km/h**
- **£300** budget

By assuming a target velocity of 20 km/h, calculated a required gear ratio **G = 9.9**.
Hence, decided to use a 3-stage transmission process, aiming for **G ~ 2.15 per stage**.



RS Order included:

- Bearings
- Taper bushing sprockets
- Chain tensioner
- Spur gears
- T-bolts
- Hex nuts

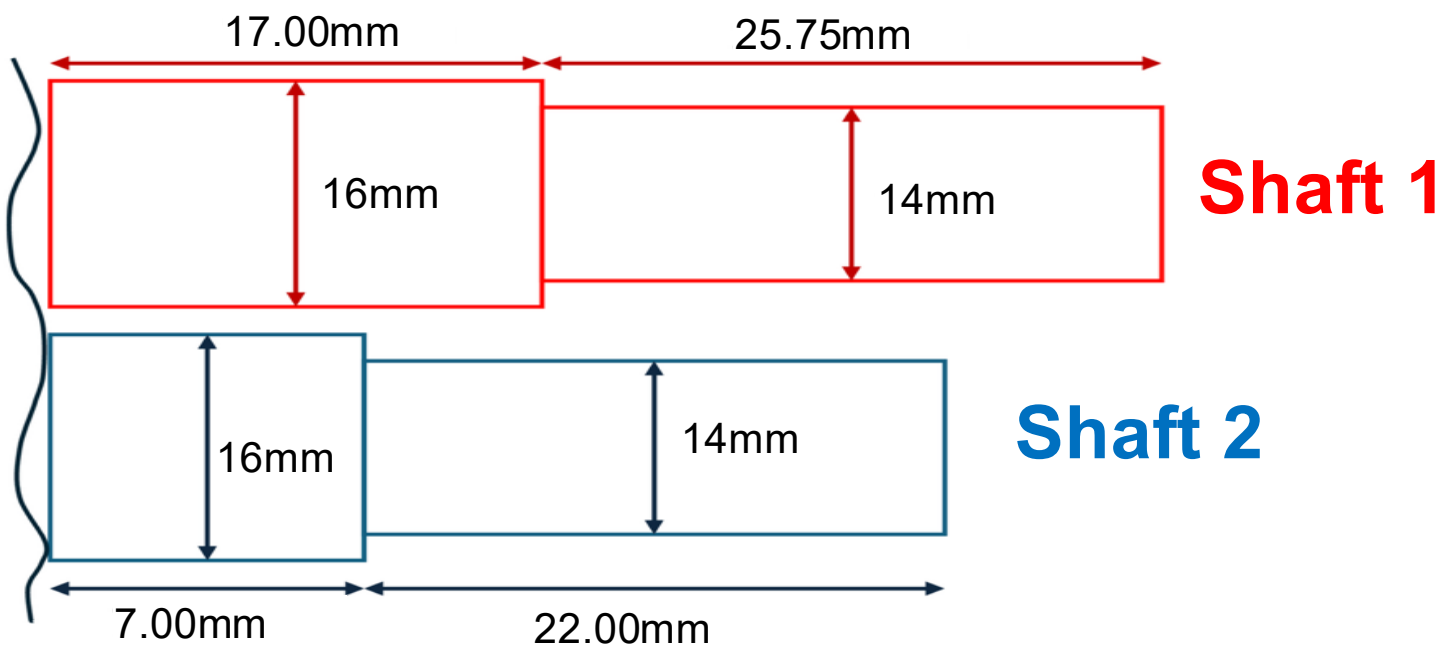
Spent £295.24

Parts purchased did not produce the exact value of G desired:
3.17 : 3 : 1 (ratio for each stage) = 9.5



Design **places motor on top of frame** (with motor plate as an adapter) to reduce distance of 1st chain, reducing chance of chain slack or derailment. But tensioner still needed for 2nd chain.

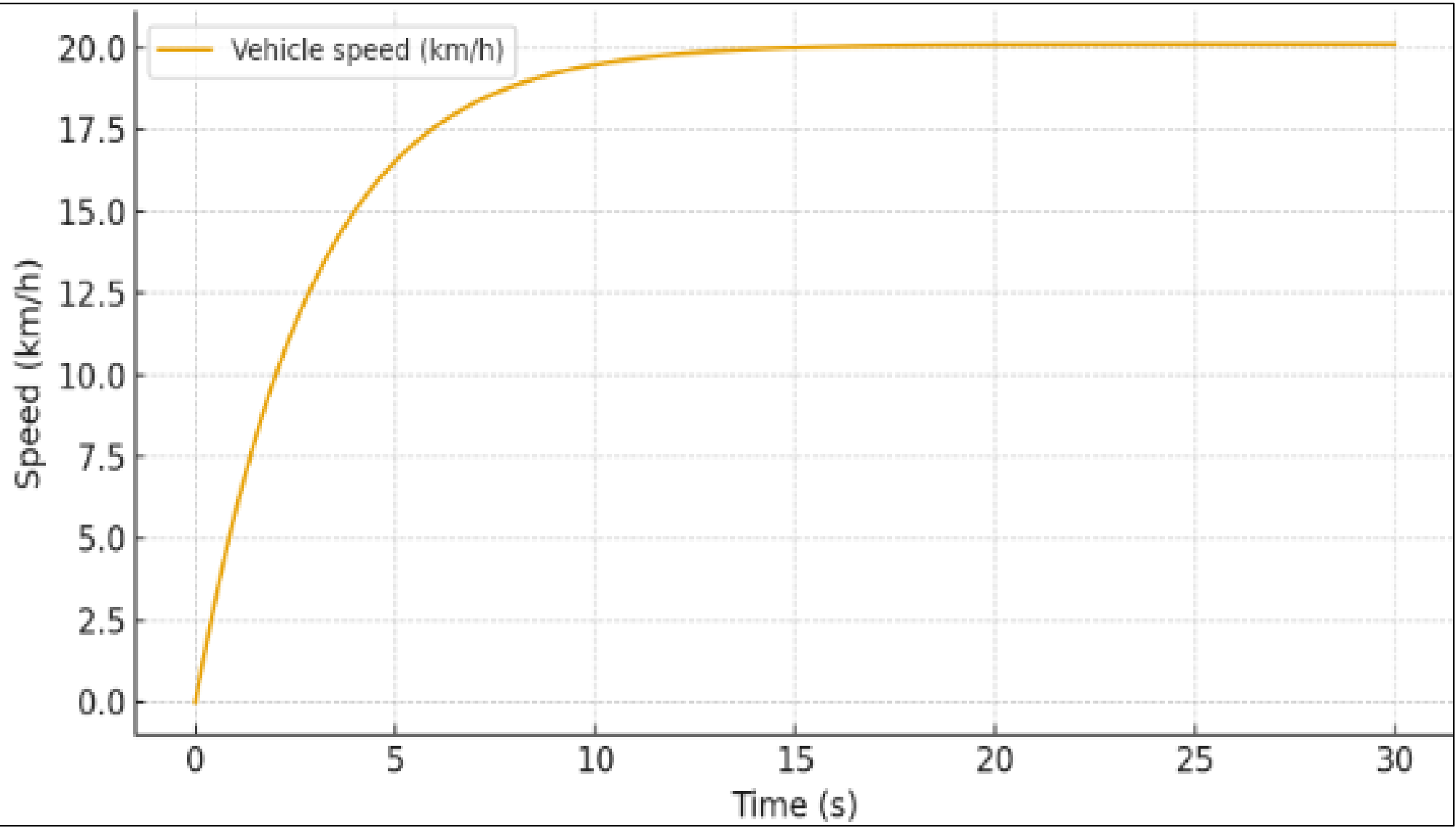
Shafts designed to be near identical. Thus, we can CNC Shaft 1 twice, then cut and turn one shaft to match desired dimensions of Shaft 2 – helping reduce manufacturing time.



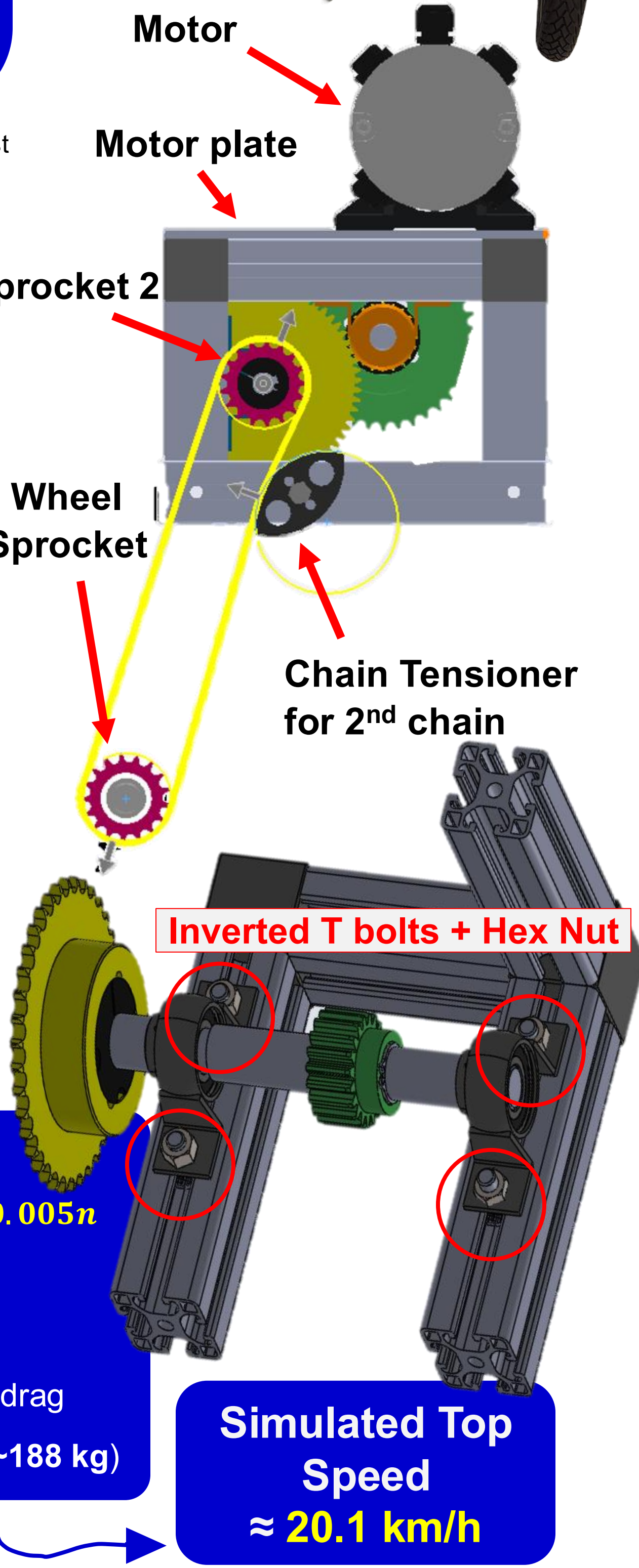
Summary of Manufactured Parts	
Part	Manufacturing Method
Shaft body	- Both shafts CNC milled - Milled keyways via slot drill - Turn Shaft 2 via lathe - Cut Shaft 2 via cutting lathe attachment
Motor plate	- Shape cut out via saw - Clearance holes via drill
Gears	- Bored via lathe - Keyways broached

Failure Assessment		
Mode of Failure	Severity	Mitigation
Incorrect Motor Direction	Medium	Trial before attaching chain
Gear misalignment	High	Tighten circlips before testing
Chain derailment	High	Remove chain links till not slack
Excessive heat	Low	Lubricate regularly

A **dynamic simulation** was carried out on Python to estimate the vehicle's acceleration profile and final running speed:



- The model incorporates:
- Motor torque curve; $T(n) = 11.5 - 0.005n$
 - Overall modification (**G = 9.5**)
 - Drivetrain efficiency (**~90%**)
 - Rolling resistance and aerodynamic drag
 - Total vehicle mass including driver (**~188 kg**)



Can conclude that F24 car will operate at desired speed output, unless chain slacks and / or derails. However, there is sufficient tensioning in design to prevent this and so should operate safely and successfully.